Symmetries in Physics — Exercise Sheet 14

The Lorentz group and spinors

Variant: Questions only

Exercise 1 Show that $\mathrm{O}(1,3)$ has exactly four connected components. Use the two homomorphisms $\det\colon \mathrm{O}(1,3) \to \{\pm 1\}$ and $\operatorname{sgn} \Lambda^0{}_0$, prove the latter is multiplicative, and identify the four cosets.

Exercise 2 For collinear boosts along a fixed axis, parametrise by rapidity χ via $\beta = \tanh \chi$. Show $B(\chi)B(\chi') = B(\chi + \chi')$, and recover the relativistic velocity-addition formula.

Exercise 3 (Thomas–Wigner rotation) Two boosts of equal rapidity χ in perpendicular directions (x then y) compose, in $\mathrm{SL}(2,\mathbb{C})$, to a pure boost followed by a rotation. Compute the rotation angle θ_W exactly, working with $A = A_2 A_1$ where $A_i = \exp(\frac{\chi}{2}\,\hat{n}_i \cdot \vec{\sigma})$.

Exercise 4 Verify that the map $\Phi\colon \mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ defined by $X \mapsto AXA^\dagger$ on Hermitian 2×2 matrices lands in $\mathrm{SO}^+(1,3)$, and compute Φ for the pure boost $A = e^{\chi\sigma_3/2}$.

Exercise 5 Realise the four-vector representation as the intertwiner $(\frac{1}{2}, \frac{1}{2}) \to$ vectors: with $E_\mu = (\mathbb{1}, \sigma_1, \sigma_2, \sigma_3)$, show $x^\mu \mapsto X = x^\mu E_\mu$ is $\mathrm{SL}(2,\mathbb{C})$-equivariant, and recover the Lorentz matrix as $\Lambda^\mu{}_\nu(A) = \frac{1}{2}\,\mathrm{tr}(E_\mu A E_\nu A^\dagger)$.

Exercise 6 (Starred) Build the Dirac matrices from Weyl blocks: $\gamma^0 = \left(\begin{smallmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{smallmatrix}\right)$, $\gamma^k = \begin{pmatrix} 0 & -\sigma_k \\ \sigma_k & 0 \end{pmatrix}$. Verify the Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{1}$ (signature $+{-}{-}{-}$) in all cases, and show the Lorentz generators $\Sigma^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu]$ split into $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. (Use the dictionary $L_a = \frac{1}{2}\epsilon_{abc}\Sigma^{bc}$, $K_a = \Sigma^{0a}$, and check Lecture 14's brackets on the blocks.)